Project: **FORCADOS SHORELINE EROSION TREND ANALYSIS TO SUPPORT THE RPA PIPELINE REBURIAL OF EXPOSED SECTION**

INTRODUCTION:

The 24" Forcados - Yokri spurline (Forcados/Yokri - OGGSRP-A), also known as the Forcados RPA pipeline connects the 24" Export Gasline running from the Forcados Yokri North Bank CPF in OML 43 in Delta state to the OGGS RPA shallow offshore platform in OML 79. The DFTP team recently discovered an exposed section of the Forcados RPA pipeline near the shoreline approximately 8 km Northwest of the Forcados estuary. Erosion due to sea level rise and incursion is suspected to be responsible.

The Geoinformation team is tasked to provide an erosion trend analysis study of the shoreline which will provide insights into the dynamics of shoreline changes over a ten-year period to help the DFTP onshore team take decisions on how best to rebury the exposed section of the pipeline.

STUDY OBJECTIVES:

The objectives of this study are:

- Identify and extract shorelines from five epochs of optical satellite images.
- Determine the rates of change of the shorelines from 2013 to all other epochs (2016, 2019, 2021 and 2024).

The methodology to achieve these objectives includes feature overlay & geometry extraction calculations, digital image processing and analysis, and visual interpretation & classifications.

STUDY AREA AND DATA AVAILABILITY:

The study area covers the 24" Forcados - Yokri spurline (Forcados/Yokri - OGGSRP-A) which is in OML 43 within the Renaissance Africa Energy Company Limited operational areas in the Niger Delta. The project area is approximately 4.01 km^2 running through the Atlantic Ocean and freshwater swamp. Availability and specification of data over the project areas are as stated in the table below:

Study area Showing the 24'' Forcados-Yokri (Forcados/Yokri-OGGSRP-A) spurline onshore landing.

S/No	Satellite Imagery
1	Available (Worldview-1, 0.5m res, Dec 2013)
2	Available (Worldview-1, 0.5m res, May 2016)
3	Available (GeoEye-1, 0.5m res, Dec 2019)
4	Available (Worldview-2, 0.5m res, Dec 2021)
5	Available (Worldview -3, 0.3m res, JUN2024)

Data Availability and Specification for the project area

DRAFT

METHODOLOGY:

IMAGE COREGISTRATION:

All the five images were first manually co-registered using the first order polynomial georeferencing with the same Ground Control Points (GCP). The cumulative RMS error is 1.

SHORELINES EXTRACTION:

The three Multispectral image epochs out of the five images available for the study were analysed for shoreline extraction by first creating Normalized Difference Vegetation Index (NDVI) images and then vectorizing the NDVI images.

$$\text{NDVI} = \frac{\text{Band 4} - \text{Band 3}}{\text{Band 4} + \text{Band 3}}$$

Where Band 3 = Reflectance in the Red band and

Band 4 = Reflectance in the Near Infra-red band

The shorelines were now extracted using the Polygon to Lines geoprocessing tool.

The two Panchromatic image epochs were visually analysed by first changing the greyscale colour ramp to a multicolour ramp and then apply a Low-pass filter to enhance the shoreline boundary.

The shoreline extracted in the study corresponds to the between the wet sand and the dry sand which is commonly used in coastline management generally.

All Analysis and technique for the shoreline extraction were carried out in ESRI's ArcGIS Pro environment.

SHORELINE RATE OF CHANGE:

The baseline and transect technique were used to calculate the shoreline rate of change. A toolbox was created that use a straight line parallel to the shoreline of the base year to generate transect lines at 50 meters interval and heights at 500 meters. These transect lines projects on to all the extracted shorelines for all the epochs. Points are generated at every intersection between the transect lines and the shorelines. Displacement measurements were taken along transect lines between two points which corresponds to locations of the shoreline at different times (year).

To understand the dynamics of erosion (erosion pattern) along the shoreline within the AOI, the shoreline is divided into three zones. Measurements will include the following.

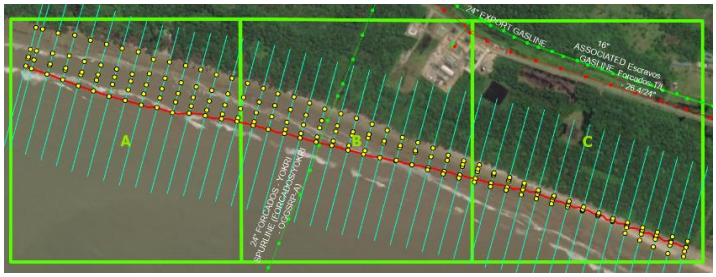
End Point Rate (EPR):

$$\text{"EPR"} = \frac{\text{"Distance between oldest and newest shoreline"}}{\text{"Time interval"}}$$

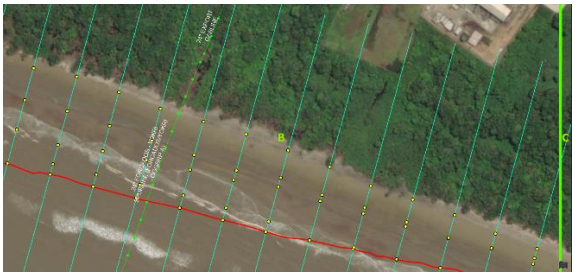
Net Shoreline Movement (NSM): Total displacement without dividing by time.

Shoreline Change Envelope (SCE): Range of shoreline positions.

Please note that field observation and measurement is required to determine the **actual** displacement of the shoreline (see recommendation(s)).



Base year Shoreline showing transect lines and the zoning polygons.

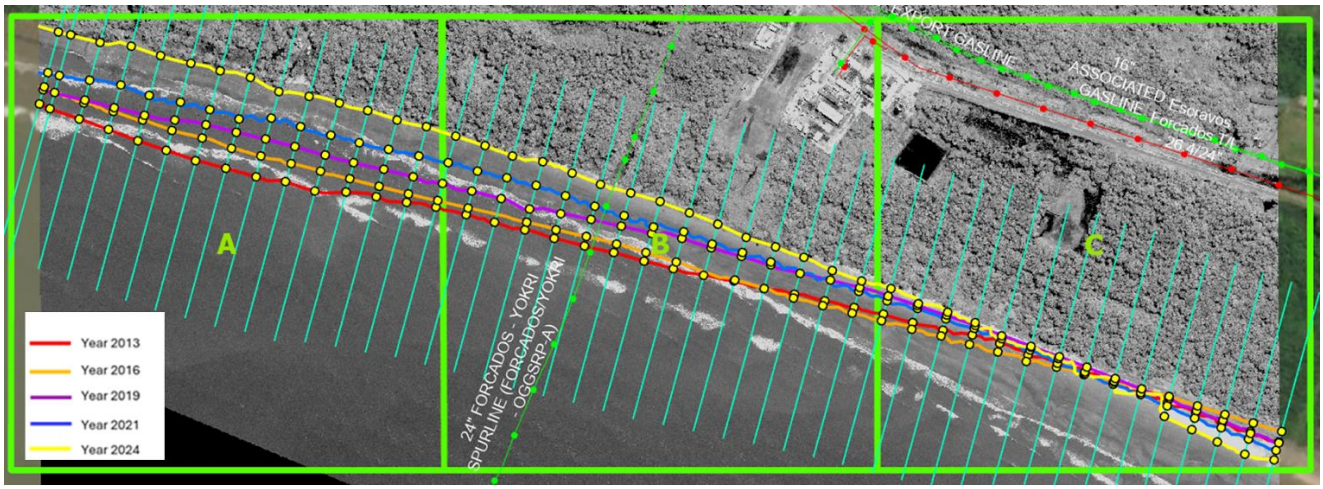


Base year Shoreline showing shoreline intersection points.

RESULTS:

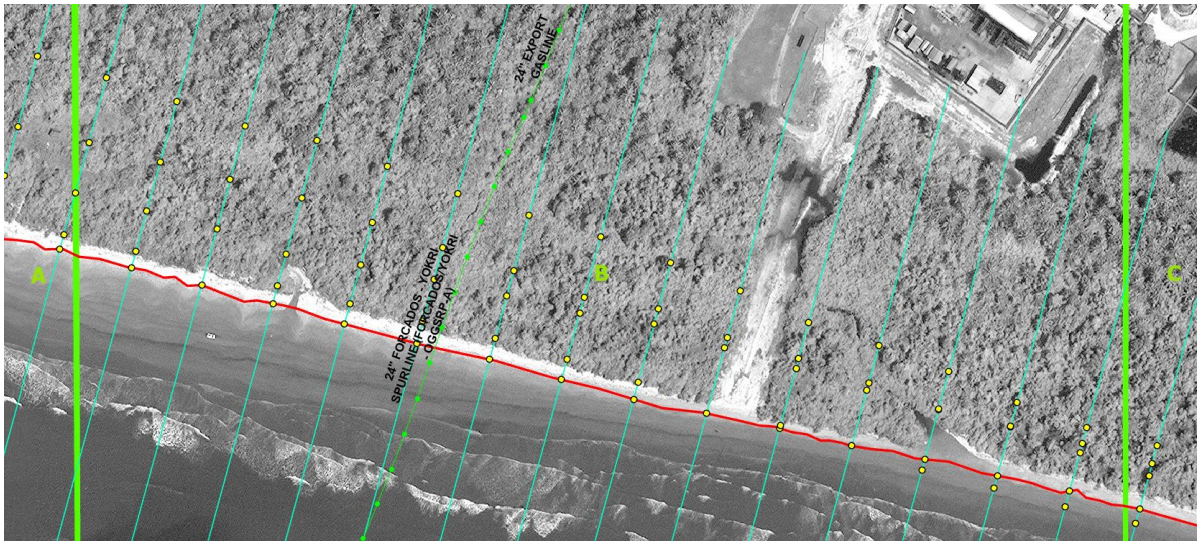
SHORELINES EXTRACTION:

The five extracted shorelines for the different epochs are presented below with their associated images.

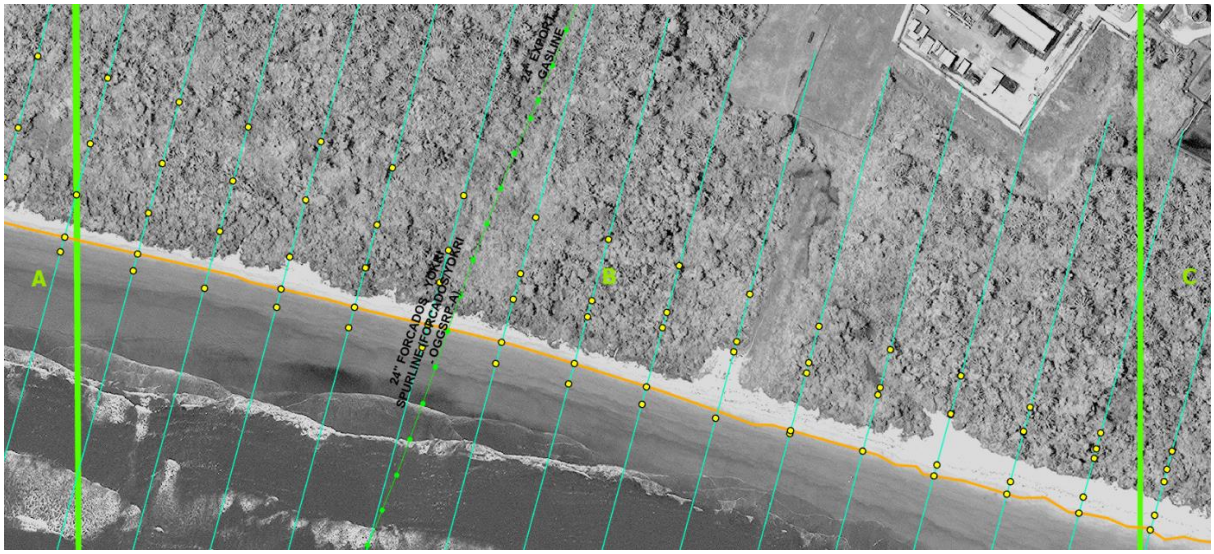


Relative positions of shoreline at different epochs.

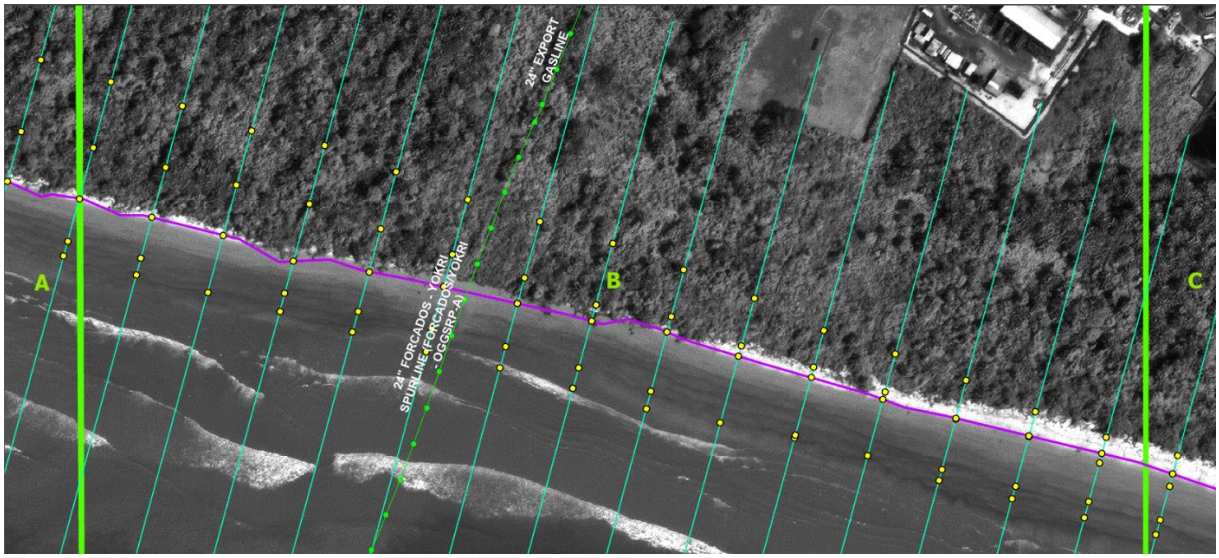
YEAR 2013:



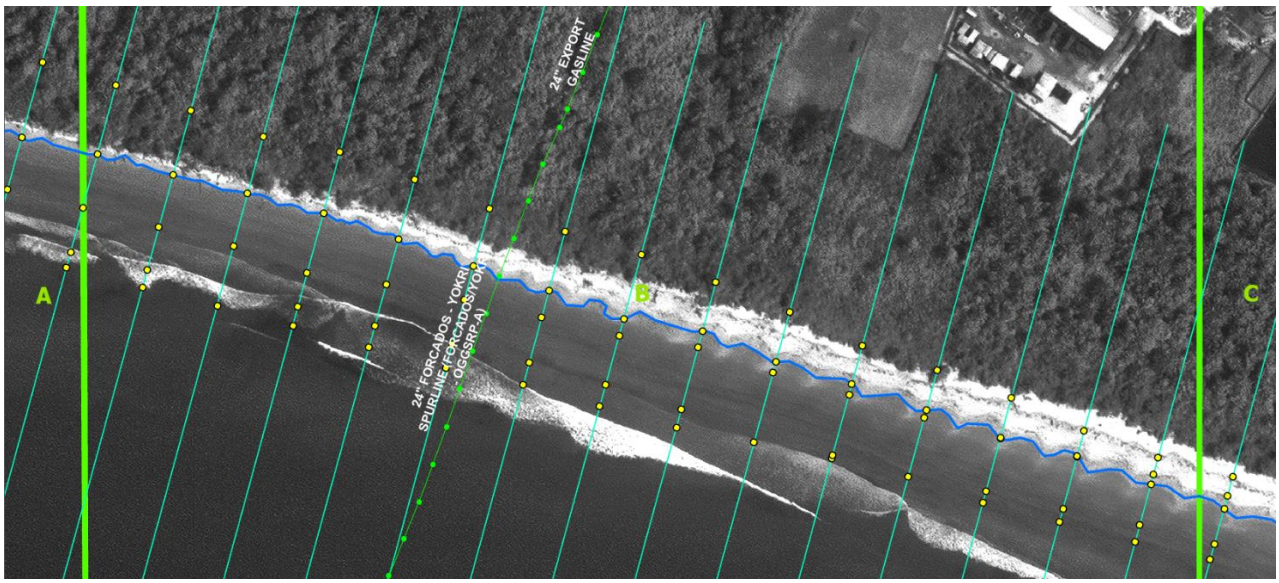
YEAR 2016:



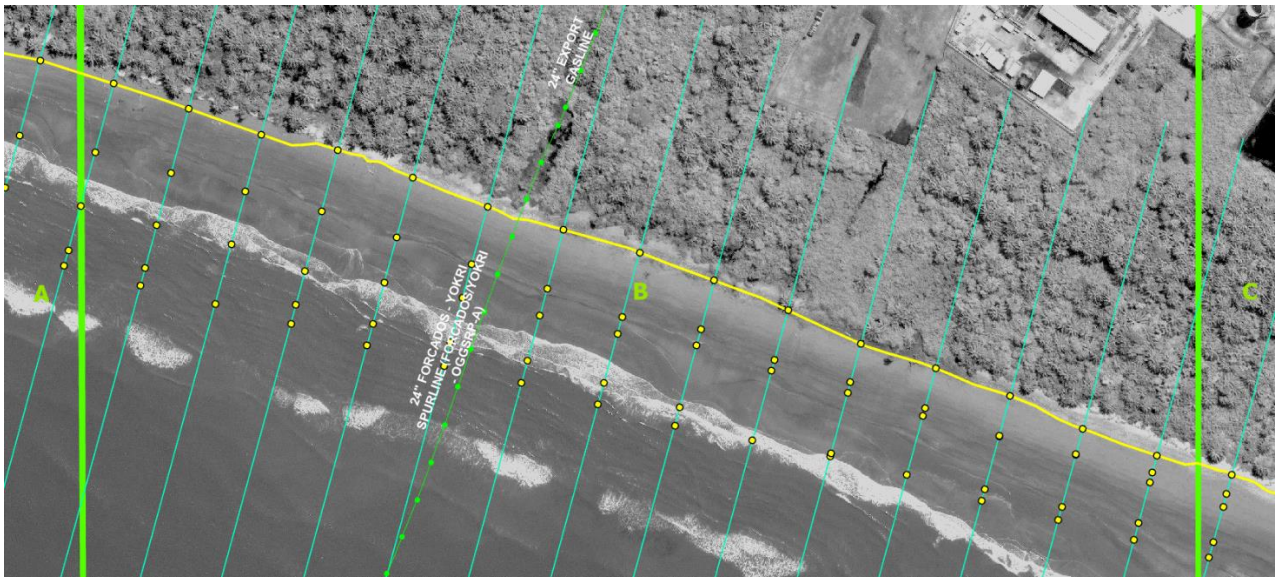
YEAR 2019:



YEAR 2021:



YEAR 2024:



SHORELINE RATE OF CHANGE:

The Net Shoreline Movement (NSM) which corresponds to the cumulative average shoreline erosion between 2013 and 2024 is approximately 202.18 meters. Between 2013 and 2016, average shoreline erosion is highest in zone "A" with approximately 27.46 meters and lowest in zone "C" with approximately 0.14 meters. The same trend persists for all other periods i.e. 2013 to 2019, 2013 to 2021 and 2013 to 2024.

			PERIODIC SHORELINE EROSION (METERS PER PERIOD)										
S/N	Point ID	Zone	2013	2013 - 2016	2013 - 2019	2013 - 2021	2013 - 2024	Point ID	2016 - 2019	Point ID	2019 - 2021	Point ID	2021 - 2024
1	A1		0	-27.89	-22.66	-52.12	-119.68	B1	5.23	C1	-29.47	D1	-67.55
2	A2		0	-27.16	-26.67	-55.81	-123.22	B2	0.49	C2	-29.14	D2	-67.42
3	A3		0	-26.47	-31.12	-65.09	-127.95	B3	-4.59	C3	-33.98	D3	-62.86
4	A4		0	-30.47	-36.8	-74.56	-140.07	B4	-6.33	C4	-37.36	D4	-65.51
5	A5		0	-33.57	-42.91	-75.24	-141.68	B5	-9.34	C5	-32.33	D5	-66.31
6	A6		0	-34.11	-50.85	-81.45	-145.55	B6	-16.73	C6	-30.6	D6	-64.10
7	A7		0	-38.01	-59.88	-85.3	-149.38	B7	-21.86	C7	-25.43	D7	-64.08
8	A8	A	0	-33.43	-60.86	-83.14	-151.66	B8	-27.43	C8	-22.28	D8	-68.52
9	A9		0	-34.14	-62.82	-98.57	-143.41	B9	-28.68	C9	-35.75	D9	-44.85
10	A10		0	-30.11	-58.4	-92.39	-143.29	B10	-28.29	C10	-33.99	D10	-50.90
11	A11		0	-33.23	-59.57	-93.32	-146.95	B11	-26.35	C11	-33.75	D11	-53.63
12	A12		0	-22.17	-49.29	-82.01	-136.26	B12	-27.12	C12	-32.98	D12	-54.26
13	A13		0	-16.94	-45	-78.44	-128.5	B13	-28.06	C13	-33.43	D13	-50.06
14	A14		0	-14.23	-43.24	-77.17	-126.18	B14	-29.01	C14	-33.93	D14	-49.01
15	A15		0	-9.99	-39.08	-73.76	118.36	B15	-29.15	C15	-34.68	D15	-44.60
Average			**0**	**-27.46**	**-45.94**	**-77.89**	**-120.36**	**Average**	**-18.48**	**Average**	**-31.94**	**Average**	**-58.24**
16	A16		0	-11.67	-39.24	-73.1	-115.14	B16	-27.56	C16	-33.87	D16	-42.04
17	A17		0	-12.52	-39.07	-73	-110.32	B17	-26.55	C17	-33.93	D17	-37.32
18	A18		0	-12.21	-33.97	-73.06	-112.99	B18	-21.75	C18	-39.1	D18	-39.93
19	A19		0	-14.23	-41.21	-70.27	-109.39	B19	-26.98	C19	-29.06	D19	-39.12
20	A20		0	-15.63	-44.15	-66.54	-103.75	B20	-28.52	C20	-22.38	D20	-37.22
21	A21		0	-14.93	-44.09	-61.36	-99.91	B21	-29.16	C21	-17.27	D21	-38.55
22	A22		0	-14.06	-46.09	-57.45	-99.18	B22	-32.02	C22	-11.11	D22	-41.8
23	A23	B	0	-12.04	-52.02	-62.47	-94.33	B23	-39.98	C23	-10.45	D23	-31.86
24	A24		0	-5.03	-45.44	-52.38	-84.76	B24	-40.4	C24	-6.95	D24	-32.37
25	A25		0	-2.02	-41.24	-48.31	-73.29	B25	-39.04	C25	-7.08	D25	-24.98
26	A26		0	-0.38	-38.51	-43.38	-69.35	B26	-38.13	C26	-4.87	D26	-25.97
27	A27		0	7.54	-34.98	-34.27	-60.88	B27	-42.52	C27	0.71	D27	-26.61
28	A28		0	8.31	-34.36	-33.96	-50.77	B28	-42.67	C28	0.4	D28	-16.81
29	A29		0	10.89	-32.99	-26.61	-44.09	B29	-43.88	C29	6.38	D29	-17.47
30	A30		0	9.95	-31.42	-23.05	-44.28	B30	-41.37	C30	8.37	D30	-21.23
Average			**0**	**-5.20**	**-39.92**	**-53.28**	**-84.83**	**Average**	**-34.70**	**Average**	**-13.35**	**Average**	**-31.55**
31	A31		0	10.22	-28.89	-23.81	-31.66	B31	-39.11	C31	5.07	D31	-7.85
32	A32		0	8.92	-24.39	-18.4	-29.98	B32	-33.32	C32	5.99	D32	-11.58
33	A33		0	16.06	-16.16	-11.26	-20.26	B33	-32.21	C33	4.89	D33	-8.99
34	A34		0	14.71	-13.63	-8.27	-8.09	B34	-28.34	C34	5.36	D34	0.18
35	A35		0	12.31	-13.73	-6.99	-12.65	B35	-26.04	C35	6.74	D35	-5.66
36	A36		0	7.03	-10.49	0.41	-7.67	B36	-17.53	C36	10.9	D36	-8.08
37	A37	C	0	1.75	-9.78	-1.38	3.41	B37	-11.53	C37	8.4	D37	4.79
38	A38		0	-0.49	-6.08	3.56	-0.38	B38	-5.58	C38	9.63	D38	-3.94
39	A39		0	-3.99	-4.14	4.69	2.13	B39	-0.15	C39	8.84	D39	-2.56
40	A40		0	-10.59	-7.48	4.17	22.28	B40	3.11	C40	11.65	D40	18.11
41	A41		0	-11.16	-3.59	8.86	28.02	B41	7.56	C41	12.46	D41	19.16
42	A42		0	-15.09	-5.01	6.9	30.5	B42	10.08	C42	11.91	D42	23.6
43	A43		0	-14.36	2.79	12.75	37.44	B43	17.15	C43	9.96	D43	24.68
44	A44		0	-17.33	0.82	13.95	29.04	B44	18.15	C44	13.13	D44	15.09
Average			**0**	**-0.14**	**-9.98**	**-1.06**	**3.01**	**Average**	**-9.84**	**Average**	**8.92**	**Average**	**4.07**
Cumulative Average			**0**	**-32.8**	**-94.9**	**-132.23**	**-202.18**	**Cumulative Average**	**63.02**	**Cumulative Average**	**-36.37**	**Cumulative Average**	**-85.72**

Shoreline displacement matrix.

The cumulative average erosion is highest between 2021 and 2024 with approximately 85.72 meters and lowest between 2013 and 2016 with approximately 32.8 meters. In zone "A", the average erosion is highest between 2021 to 2024 and lowest in 2016 and 2019 with approximately 18.48 meters but in zone "B", the erosion average is highest between 2016 and 2019 with approximately 34.7 meters and lowest between 2013 and 2016 with approximately 5.20 meters. In zone "C", where accretion was observed between 2019 and 2021 (8.92 meters) and between 2021 and 2024 (4.07 meters). The highest average erosion recorded is 9.84 meters between 2016 and 2019.

The End Point Rate (EDR) is the rate of shoreline movement between two shoreline positions (the earliest and the latest) along a transect over a given period. Below is the breakdown of EDR for the area of interest.

CUMULATIVE ANNUAL RATE OF EROSION (METERS PER TIME INTERVAL)								
2013	**2013 - 2016**	**2016 - 2019**	**2019 - 2021**	**2021 - 2024**		**2013 - 2019**	**2013 - 2021**	**2013 - 2024**
0	-10.93	-21.01	-18.19	-28.57		-15.82	-16.53	-18.38

Cumulative annual rate of erosion

Zone	ANNUAL RATE OF EROSION (METERS PER TIME INTERVAL)								
	2013	**2013 - 2016**	**2016 - 2019**	**2019 - 2021**	**2021 - 2024**		**2013 - 2019**	**2013 - 2021**	**2013 - 2024**
A	0	-9.15	-6.16	-15.97	-19.41		-7.66	-9.74	-10.94
B	0	-1.73	-11.57	-6.68	-10.52		-6.65	-6.66	7.71
C	0	-0.04	-3.28	4.46	1.36		-1.66	-0.13	-0.27

Annual rate of erosion by zones.

The Shoreline Change Envelope (SCE) shows the variability or range of shoreline positions over a given period. It doesn't show the direction of erosion or accretion but only the range of movement. Below is the SCE of the shoreline of interest.

PERIOD	MAXIMUM DISTANCE LANDWARD (METERS)	MAXIMUM DISTANCE SEAWARD (METERS)
2013	0	0
2013 - 2016	-38.01	16.06
2016 - 2019	-43.88	18.15
2019 - 2021	-37.36	8.37
2021 - 2024	-68.52	24.68
2013 - 2019	-62.82	2.79
2013 - 2021	-98.57	13.95
2013 - 2024	-151.66	37.44

The shoreline change envelope showing maximum displacement landward and seaward.

CONCLUSIONS & RECOMMENDATION:

CONCLUSIONS:

The study successfully deployed remote sensing and geospatial technologies to identify and extract shorelines from five epochs of optical satellite images and also determine the rates of change of the shorelines from 2013 to all other epochs (2016, 2019, 2021 and 2024), which will enable the DFTP Onshore team plan effectively and take well-informed management decisions on the reburial of the exposed pipeline.

The total area of study is approximately 4.01 km^2. The five extracted shorelines have been zoned into three (A, B and C) and analyzed for Net Shoreline Movement (NSM) which shows that the cumulative average erosion is highest between 2021 and 2024 with approximately 85.72 meters and lowest between 2013 and 2016 with approximately 32.8 meters. In zone "A", the average erosion is highest between 2021 to 2024 and lowest in 2016 and 2019 with approximately 18.48 meters but in zone "B", the erosion average is highest between 2016 and 2019 with approximately 34.7 meters and lowest between 2013 and 2016 with approximately 5.20 meters. In zone "C", where accretion was observed between 2019 and 2021 (8.92 meters) and between 2021 and 2024 (4.07 meters). The highest average erosion recorded is 9.84 meters between 2016 and 2019. The breakdown by zones is shown below.

The End Point Rate (EDR) computations show a general increment from 10.92 meters between the period 2013 to 2016 to 21.01 meters between 2016 to 2019 but reduced to 18.19 meters between 2019 and 2021 and then increased again to 28.57 meters between 2021 to 2024.

Zone	ANNUAL RATE OF EROSION (METERS PER TIME INTERVAL)								
	2013	2013 - 2016	2016 - 2019	2019 - 2021	2021 - 2024		2013 - 2019	2013 - 2021	2013 - 2024
A	0	-9.15	-6.16	-15.97	-19.41		-7.66	-9.74	-10.94
B	0	-1.73	-11.57	-6.68	-10.52		-6.65	-6.66	7.71
C	0	-0.04	-3.28	4.46	1.36		-1.66	-0.13	-0.27

The general erosion trend shows relatively higher erosional activities in zone "A" than zone "B". The exposed section of the Forcados RPA pipeline is within Zone "B" and is potentially at greater risk of further exposure due to its proximity (less than 20 meters from on-screen measurement) considering the 10.52 meters annual erosion in zone "B". The shoreline in Zone "C" experienced the least erosion and is currently experiencing accretion at the west end of the shoreline. Finally, the Shoreline Change Envelope (SCE) which is the maximum displacements landward and seaward for the shoreline under study is 151.66 meters and 37.44 meters respectively.

This study finds the 24" Forcados - Yokri spurline (Forcados/Yokri - OGGSRP-A), also known as the Forcados RPA pipeline connects the 24" Export Gasline running from the Forcados Yokri North Bank CPF in OML 43 in Delta state to the OGGS RPA shallow offshore platform in OML 79 seriously threatened by erosion caused by the ocean surge along the coastline within the area of study.

RECOMMENDATIONS:

1. The actual shoreline displacement measurements and, or status should be verified during ground truthing as this study was done using satellite image data acquired in 2013, 2016, 2019, 2021 and 2024.

Authored by:	Tunde Shebolatan (Remote Sensing Analyst, IUC/G/UUDGG)
Reviewed by:	Olumuyiwa Abimbola (Geoinformation Consultant, IUC/G/UUDGG)
	Awotiwon Olusola (Project Surveyor, IUC/G/UUDGP)
	Tariah Victor (Geoinformation Analyst, IUC/G/UUDGG)
Approved by:	Deborah Uraechu (Team Lead, Geoinformation Management, IUC/G/UUDGG)